

Ahmed Fathy

Software Engineer

ahmedfathi20044002@gmail.com | +20 155 285 1443 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Haram, Giza, Egypt 12556

PROFESSIONAL SUMMARY

Detail-oriented Software Engineer with expertise in building full-stack web applications, RESTful APIs, and scalable backend architectures. Proven ability to optimize application performance and deploy robust solutions using modern technologies such as React, Node.js, and SQL databases.

EXPERIENCE

- **Al Nada Scientific Office** | *Software Developer (Part-time)* Cairo, Egypt | June 2022 – 2025
 - **ERP System:** Developed a local desktop application (packaged with Electron) managing customers, suppliers, inventory, and sales records using React.js, Express.js, and PostgreSQL.
 - **Corporate Web Solution:** Built the official company website integrated with an e-commerce platform designed to seamlessly manage and showcase sponsors, partners, services, and dynamic product catalogs.

PROJECTS

- **Hankers (Social Platform)** | *Full-Stack Engineering* Source Code | 2025
 - **Architecture:** Built scalable social platform using Microservices principles, React.js, and TypeScript.
 - **Frontend:** Implemented responsive UI design and asynchronous API integration via AJAX.
 - **DevOps:** Established CI/CD pipelines and containerized services using Docker for consistent deployment.
- **TCCD Official Website & Management System** | *Full-Stack Development — React, Typescript* Source Code | 2026
 - **Impact:** Developed a comprehensive student activity platform that successfully scaled to support and manage over 20,000 active users.
 - **Core Features:** Engineered a robust event and judging system that handles registration slots, waiting lists, and distinct role-based access for members, standard users, and corporate entities.
 - **Administration:** Built an advanced admin dashboard equipped with user-facing analytics to streamline the management of events, user data, and company profiles.
- **Interactive Orrery – NASA Space Apps Hackathon** | *3D Space Simulation, React.js* Source Code | Oct 2024
 - **Hackathon Project:** Developed during NASA Space Apps Challenge 2024, simulating orbital mechanics and near-Earth object trajectories using React.js and Three.js.
- **Ocean Adventure: Grand Tide (3D Survival RPG)** | *C++, OpenGL, GLSL* Source Code | 2026
 - **Graphics & Rendering:** Built a custom 3D rendering engine utilizing a Forward Renderer with transparent object sorting, Phong shading models, and post-processing effects.
 - **Shader Programming:** Implemented custom GLSL shaders, including a dynamic water shader featuring Fresnel reflections, refraction mapping, and summed sine wave displacement.
 - **Architecture & Logic:** Engineered a lightweight Entity-Component-System (ECS) to handle 3D meshes, bounding-box collision detection, ray-casting combat, and procedural entity generation.
- **Image Restoration Pipeline** | *Deep Learning & Computer Vision, Python* Source Code | 2024
 - **Pipeline:** Constructed end-to-end restoration pipeline: Denoising, Deblurring, and Inpainting.
 - **Algorithms:** Applied Telea (FMM) inpainting and deep learning models to restore damaged historical photographs.
- **NexaMart** | *E-commerce Solution, Full-Stack* Source Code | 2025
 - **Stack:** Next.js (TypeScript), Redux Toolkit, Laravel 11, and MySQL.
 - **Functionality:** Implemented secure authentication (Sanctum), shopping cart logic, real-time order processing, and administrative analytics dashboard.

EDUCATION

- **Cairo University, Faculty of Engineering** Giza, Egypt
 - *B.Sc. in Computer Engineering; GPA: 3.6* 2022

STUDENT ACTIVITIES

- **IEEE** | *UI/UX Designer - Frontend Team Leader* *Cairo | Oct 2025 – Present*
 - **Portfolio Platform Development:** Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- **Technical Center for Career Development (TCCD)** | *Frontend Developer* *Cairo | June 2025 – Present*
 - **Portfolio Platform Development:** Developed comprehensive platform with event pages, admin dashboard for event creation, role-based access for instructors and students, location management, event galleries, and statistical analysis pages.
 - **HR Management & Judging System:** Built HR system managing team attendance with custom form builder and reviewer functionality similar to Google Forms, alongside event judging system for evaluating and assessing teams applying to events.
- **Physics Helping District (PhD)** | *Team Leader* *Cairo | June 2022 – 2024*
 - **Mentorship:** Led peer tutoring initiative in electronics, achieving measurable academic improvement.
- **NASA Space Apps Challenge** | *Participant – Project: "Orrery"* *Cairo | Oct 2024*
 - **Orrery App:** Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

TECHNICAL SKILLS

- **Programming Languages:** C++ Programming, Python, JavaScript, TypeScript, SQL, PHP, Java, Assembly Language, OpenGL, GLSL
- **Core Computing Concepts:** Data Structures, Algorithms, Object-Oriented Programming, Design Patterns, Operating Systems, System Architecture, Cryptography
- **Systems Programming:** Unix/Linux Development, Multithreading, Memory Management, Network Socket Programming
- **Backend Development & DevOps:** Node.js Runtime, Express Framework, Laravel, RESTful APIs, Docker Containers, Git Version Control, CI/CD Pipelines